RENTASALUD: a web-based interactive atlas of social inequalities and life expectancy in Andalusia (southern Spain)

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Abstract

Background Interactive tools linking income and life expectancy at fine geographical scales are scarce for southern Europe. **Methods** We linked INE census-section income data (48,262 section-years, 2015–2022) with BDLPA cohort life tables (~637,000 individuals, follow-up 2011–2023) for Andalusia, Ceuta, and Melilla. **Results** Life expectancy was 79.5 (men) and 84.4 years (women). Tumours yielded the largest potential gain for men (3.93 years) and heart disease for women (2.89 years). The provincial income–EV correlation was weak (r=0.28, P=0.43), rising to r=0.52 with Ceuta and Melilla excluded. Provincial EV ranged from 79.4 (Almería) to 82.7 years (Córdoba). **Conclusion** Provincial aggregation attenuates the income–life expectancy gradient; neighbourhood-level tools like RENTASALUD are needed to detect inequalities. **Keywords:** health inequalities, life expectancy, spatial epidemiology, interactive atlas, Andalusia * GRG and PMG contributed equally as co-first authors.

1 What is already known on this topic

- People in Spain's most deprived neighbourhoods live 3–4 years less than those in the wealthiest.
- Southern Europe has smaller health inequalities than Northern or Eastern Europe.

- Interactive tools linking neighbourhood income and life expectancy are rare.

2 What this study adds

- RENTASALUD is an open source interactive atlas of census-section income and life expectancy for Andalusia, Ceuta, and Melilla.
- At province level the income–EV correlation is weak (r=0.28) and doubles when income-outlier cities are excluded (r=0.52).
- Cancer and cardiovascular disease account for 7.1 years of potential EV gain in men and 5.7 in women.
- A built-in AI assistant lets anyone query the data in plain language.

3 Introduction

In Spain, people living in the most deprived neighbourhoods die 3 to 4 years earlier than those in the wealthiest [@redondo2022]. The gradient is steep, and it varies by region. Andalusia, Spain's largest and one of its poorest regions, is home to some 8.5 million people. Yet few interactive tools allow researchers, policymakers, or the public to explore how income and life expectancy relate at the neighbourhood scale. Most available maps cover entire provinces, losing the within-area variation that drives the gradient.

RENTASALUD was built to fill that gap: a free, open source web atlas that combines detailed census-section data (INE, 2015–2022) with cohort life tables from the Longitudinal Database of the Andalusian Population (BDLPA, ~637,000 individuals followed from the 2011 census through 2023). The analysis covers the eight Andalusian provinces plus the autonomous cities of Ceuta and Melilla. We estimated life expectancy by province, sex, and cause of death, examined the income–longevity association, and deployed everything as an interactive Shiny application with an AI-powered natural-language interface.

4 Methods

Income and demographic data come from the INE *Atlas de Distribución de Renta de los Hogares* [@ine_adrh_metodologia], providing 48,262 census-section observations (2015–2022) across ten territories. Mortality data come from the BDLPA [@ieca_bdlpa_metodologia], a 10% random sample of the 2011 Andalusian census with follow-up through December 2023. Causes of death were grouped into ten broad categories.

We used standard abridged life table methods [@chiang1968] with 5-year age bands (0–4 through 85–89, 90+) to estimate life expectancy at birth by sex. Cause-deleted life tables [@beltran-sanchez2008] quantified potential gains from eliminating each cause, using Chiang's proportional-risk adjustment. Twenty life tables were constructed (10 territories × 2 sexes). Census-tract estimation was not feasible because BDLPA identifiers carry no sub-provincial geographic code.

We examined the province-level association between mean income and life expectancy using Pearson and Spearman correlations. As a sensitivity check we repeated the analysis excluding Ceuta and Melilla, two autonomous cities whose economies rely on cross-border trade and public-sector employment, making them structurally different from the Andalusian provinces. The study was approved by the University of Granada Ethics Committee (SOCESVIDA, SICEIA-2025-000814).

5 Results

Life expectancy at birth was 79.5 years for men and 84.4 for women, a gap of nearly 5 years. Provincial EV ranged from 77.8 years (Almería, men) to 80.1

(Córdoba and Cádiz, men) and from 80.9 (Almería, women) to 85.3 (Córdoba, women). The mean provincial spread was 3.3 years (79.4 in Almería to 82.7 in Córdoba).

Eliminating malignant tumours would add 3.93 years to male life expectancy at birth and 2.83 to female (Table 1). For women, heart and circulatory disease had the largest impact (2.89 years). Together, tumours and cardiovascular disease accounted for 7.1 years of potential gain in men and 5.7 in women. Men lost 1.1 more years to cancer than women, consistent with historically higher smoking rates in Spanish men.

Table 1: Years gained at birth if each cause were eliminated.

Cause	Men (years)	Women (years)
Malignant tumours	3.93	2.83
Heart and circulatory diseases	3.15	2.89
Respiratory diseases	1.13	0.69
External causes (accidents, suicide)	0.67	0.33
Digestive diseases	0.57	0.48
Nervous system diseases	0.50	0.57
Infectious diseases	0.41	0.35
Diabetes and endocrine diseases	0.28	0.32
Alcohol-related causes	0.07	0.02
Other causes	1.13	1.19

Province-level income and life expectancy were weakly correlated (r=0.28, 95% CI −0.42 to 0.77, P=0.43; Spearman =0.24, P=0.51). Ceuta (€16,899, EV 82.2) and Melilla (€15,443, EV 82.4) are income outliers: their median incomes are well above the Andalusian range (€12,699–€14,338), but their life expectancy sits in the middle, weakening the aggregate association. Excluding them raises the correlation to r=0.52 (P=0.19, R^2=0.29).

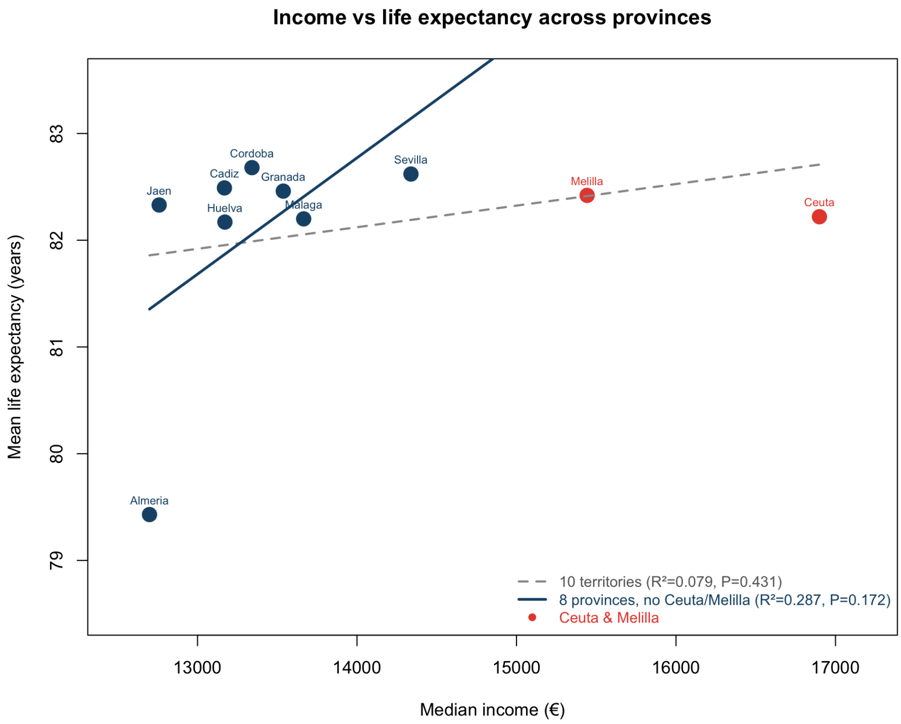

Figure 1: Income vs life expectancy across provinces. Blue: Andalusian provinces. Red: Ceuta and Melilla (income outliers). Solid line: fit without Ceuta/Melilla (R^2=0.29, P=0.17). Dashed line: all ten territories (R^2=0.08, P=0.43).

The 3.3-year provincial gradient runs in the expected direction, but the statistical signal is imprecise with ten data points. Redondo-Sánchez et al. found a much steeper gradient at census-tract level (3.8 years for men, 3.2 for women) [@redondo2022], suggesting that within-province disparities likely exceed what we can detect here. The results are available through an interactive R Shiny application at watzile.shinyapps.io/RENTA with all analysis code at github.com/migariane/RENTASALUD.

6 Discussion

The province-level income–EV correlation is weak mainly for two reasons: aggregating thousands of census tracts into ten units shrinks variance, and Ceuta and Melilla act as leverage points with high incomes but mid-range life expectancy. Removing them triples the explained variance. The 3.3-year provincial spread in life expectancy is real, but its statistical expression is noisy at n=10.

This study links individual follow-up on nearly 637,000 people over 12 years to fine-grained socioeconomic data. The main limitation is geographic: BDLPA person identifiers contain no sub-provincial code, restricting EV estimation to the province level. Future linkage to the municipal register (Padrón Continuo) would enable small-area Bayesian estimation. Other limitations include constant-mortality assumptions, small cell counts at extreme ages, and the standard caveat that cause-deleted life tables treat causes as independent.

7 Acknowledgements

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8 Data Availability

Analysis code: github.com/migariane/RENTASALUD. Shiny application: watzile.shinyapps.io/RENTA. BDLPA microdata available from IECA on request.